



London Ambulance Service



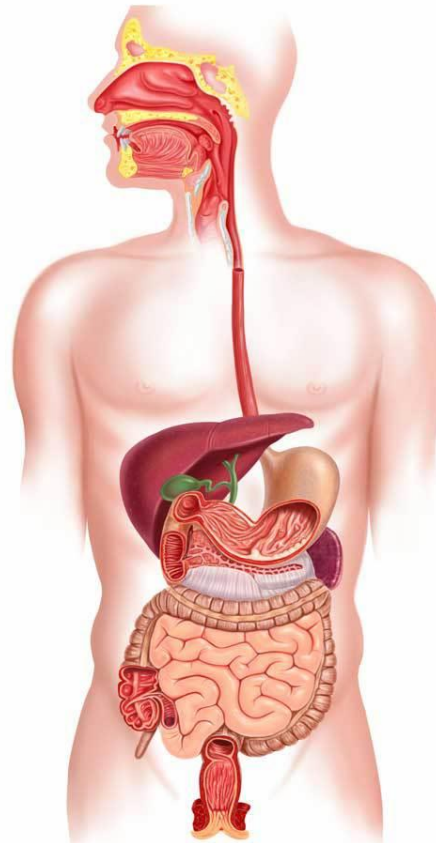
NHS Trust

Anatomy and Physiology – Component 2 + 5

The Digestive System



Digestive System - Component 2+5





Digestive System - Component 2+5

Objective

- *Explain the structure of the Digestive System*
- *Explain the functions of the Digestive System*

Digestive System - Component 2+5

Why do we need food ?

Provides nutrients for:

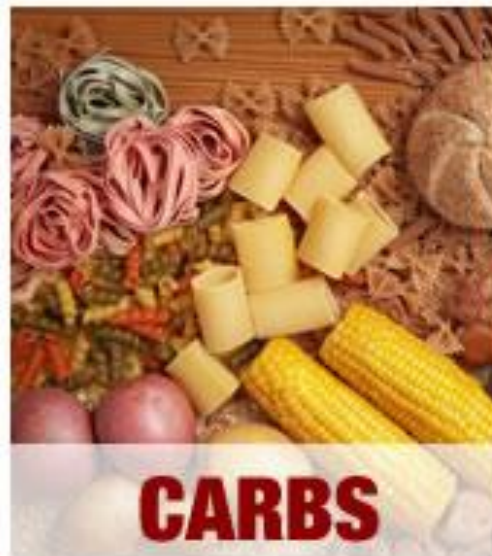
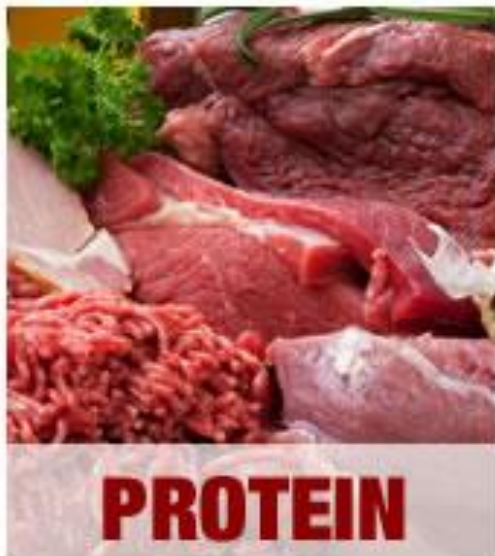
- Use as energy for cells
- Storage of energy
- Production of heat
- Growth and repair



Digestive System - Component 2+5

Essential Nutrients

These can be broken down into three basic food groups



In addition, minerals and vitamins
are also absorbed



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Components

The digestive system is the collective name used to describe:

- **Digestive processes**
- **Alimentary canal or GI tract**
- **Accessory organs of digestion**



Digestive System - Component 2+5

Digestive Processes

Food is both physically and chemically complex in composition and requires **5 stages of digestion** before it can be absorbed into the bloodstream and lymph

Ingestion → Propulsion → Digestion → Absorption → Elimination



Digestive System - Component 2+5

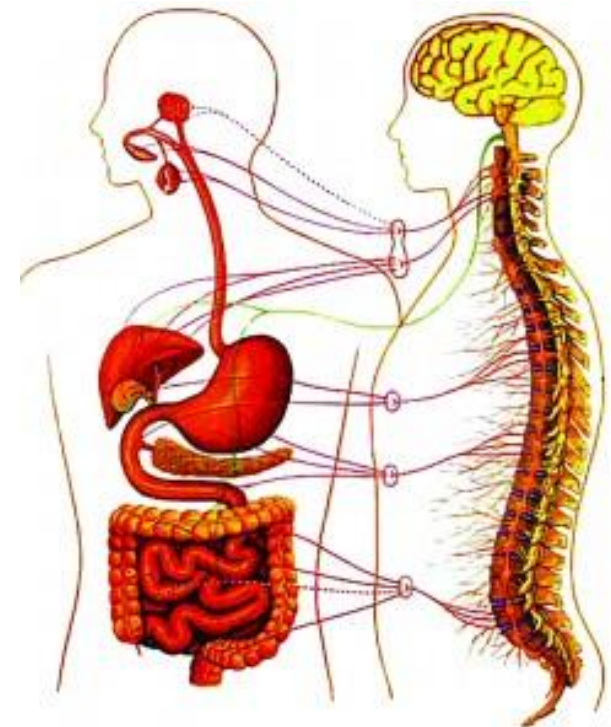
Stages of the Digestive Process

- 1. Ingestion** — Taking of Food into Alimentary Canal
- 2. Propulsion** — Mixing/moving of food along canal
- 3. Digestion** — Mechanical & Chemical Digestion
- 4. Absorption** — Digested substances from food enter blood/lymph vessels
- 5. Elimination** — Excretion of waste

Digestive System - Component 2+5

Nerve Supply to GI Tract

- Supply via ANS(Involuntary)
Parasympathetic – ‘Rest & Digest’ – **increases** peristalsis and gland secretion (Gastric Juice + Enzymes)
- **Sympathetic** ‘Fight or Flight’ – **reduces** peristalsis and secretion
- Influence each other – Activity like eating will increase Parasympathetic activity





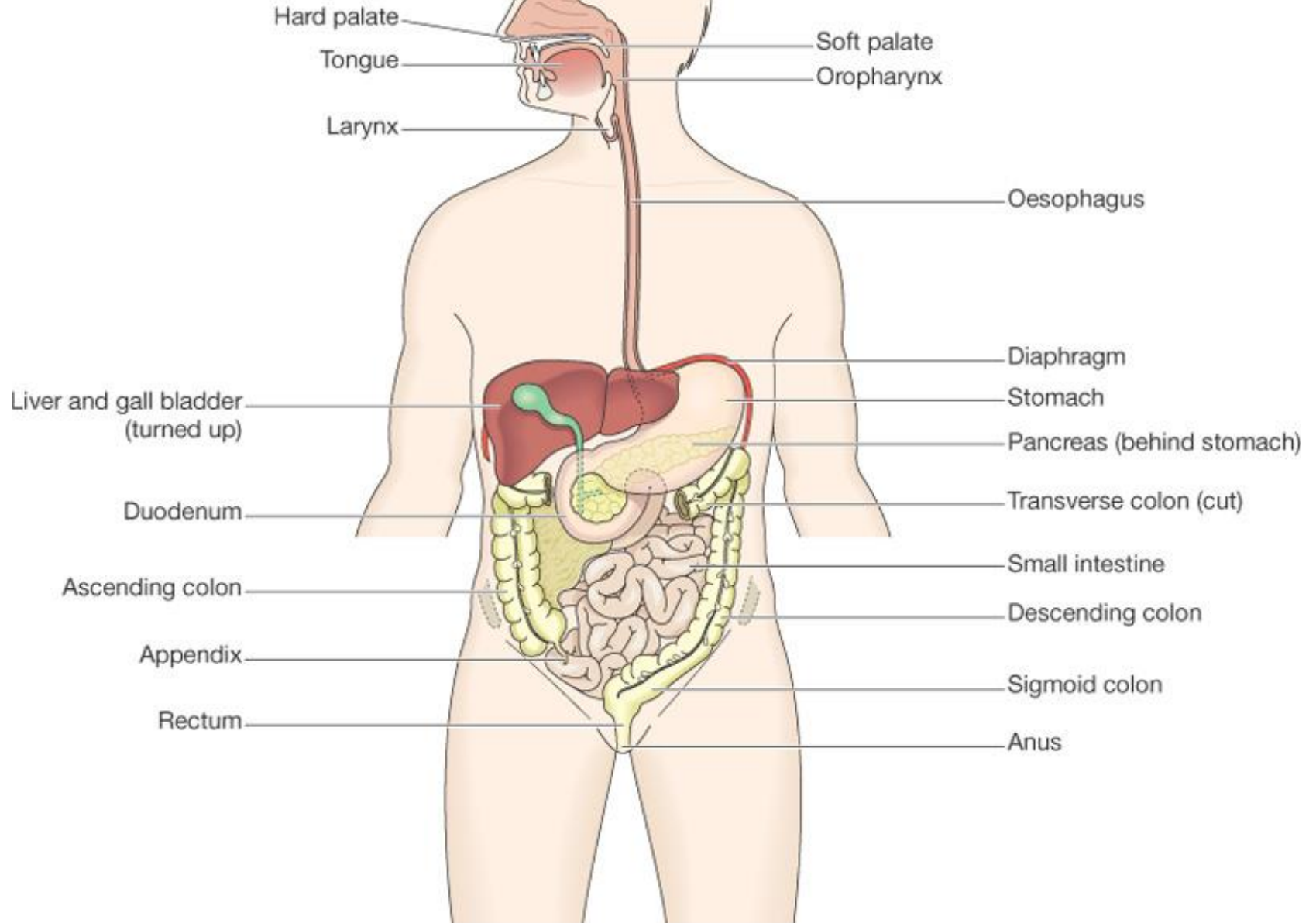
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Alimentary Tract or Canal

Also termed the Gastro - Intestinal or GI tract

Actively transports the flow of food through the whole of the digestive process

Alimentary Tract

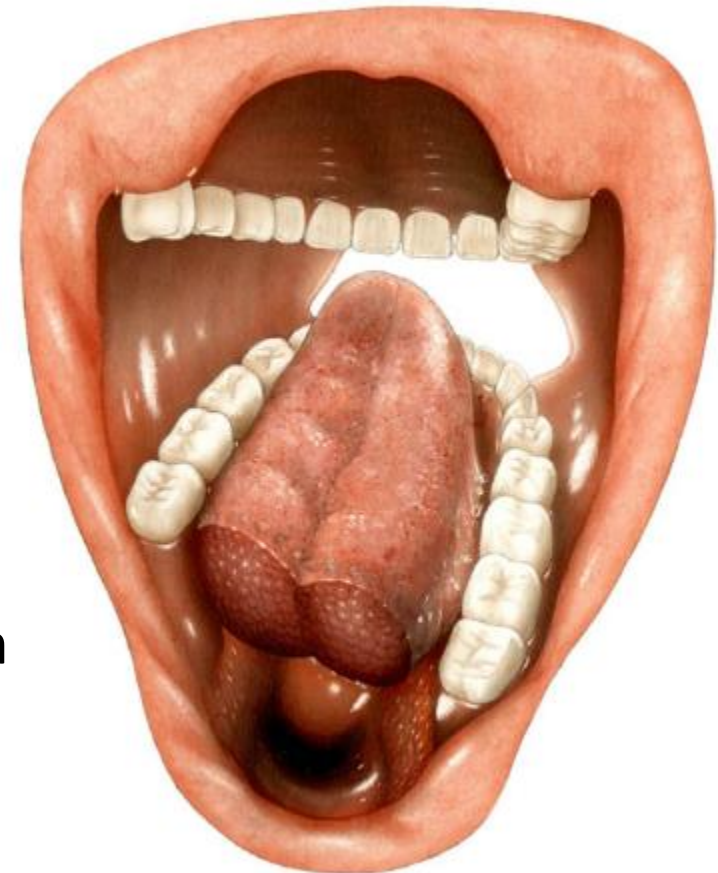


Digestive System - Component 2+5

The Mouth

The mouth also referred to as the oral cavity contains:

- The Tongue
 - Propels food bolus into oesophagus
- The Teeth
 - Chewing breaks down food (mechanical digestion)
- Salivary Glands
 - Saliva contains enzymes which begin chemical digestion



Digestive System - Component 2+5

Pharynx

The pharynx (throat) is an expanded muscular tube lying behind the nose and mouth

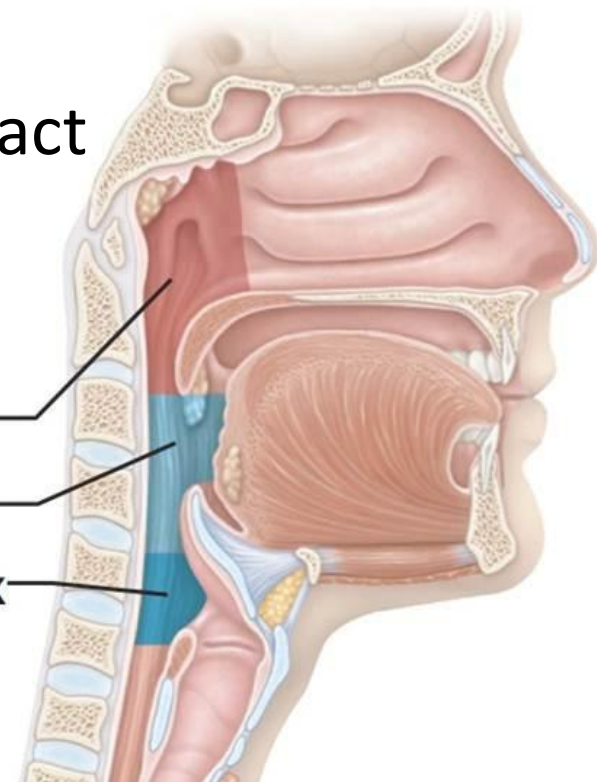
- It is shared with the respiratory tract
- *Divided into 3 parts:*

Pharynx

Nasopharynx

Oropharynx

Laryngopharynx

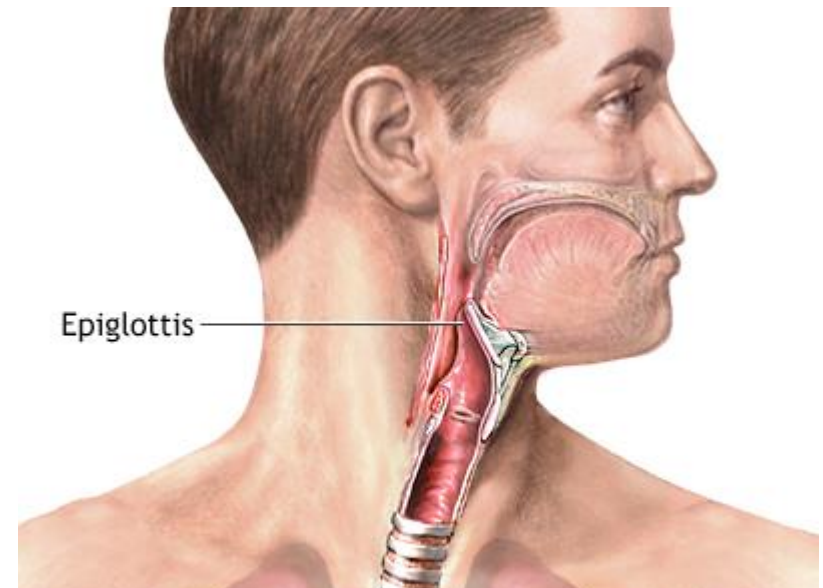


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Epiglottis

A leaf - shaped plate of yellow fibro - elastic cartilage located at the entrance of the larynx

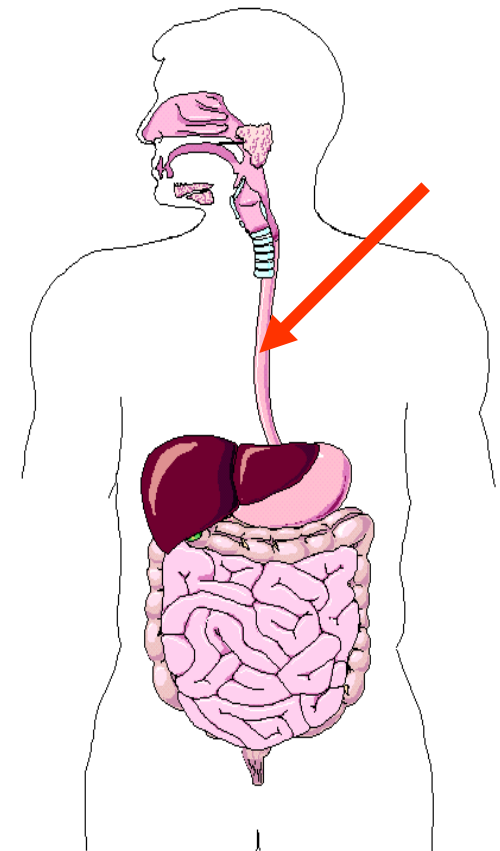
It covers the trachea whilst swallowing to stop food/fluids entering the lungs



Digestive System - Component 2+5

Oesophagus

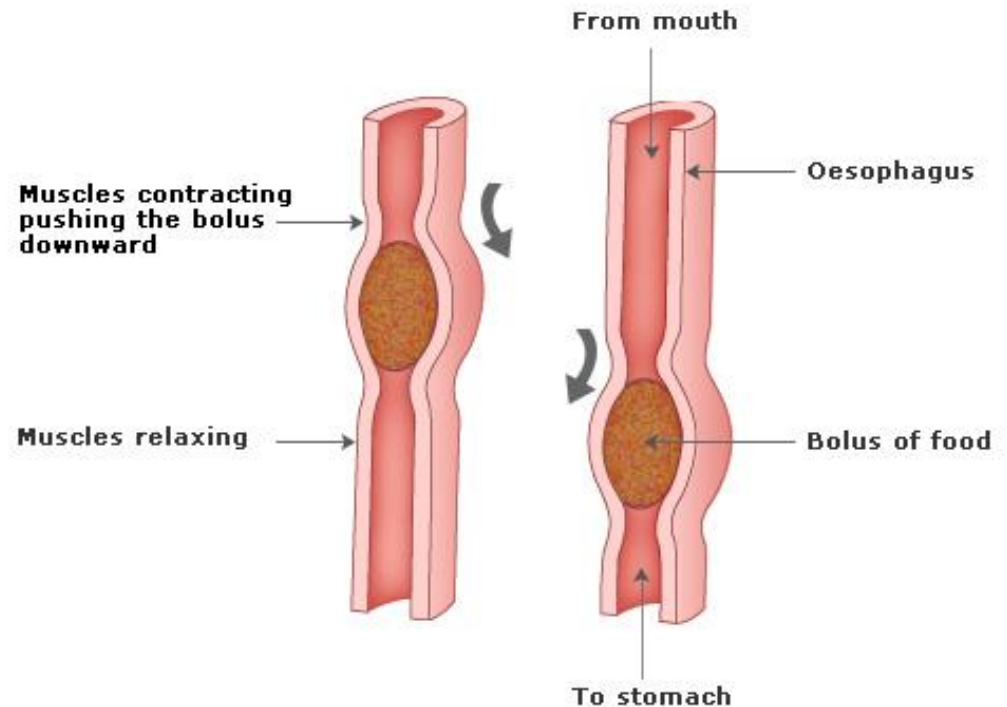
- The Oesophagus or gullet is a collapsible, muscular tube descending from the pharynx to the stomach
- It is approx. 25 cm long and 2cm in diameter in an adult
- Peristalsis moves food down
- Made up of smooth muscle, elastic tissue and a mucous membrane
- Contains Sphincters which act as valves



Digestive System - Component 2+5

Peristalsis

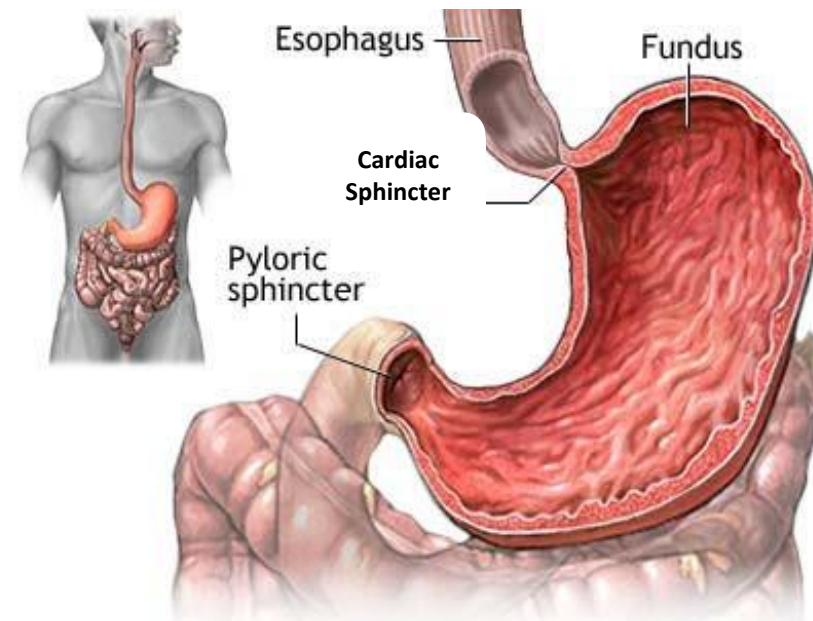
- The movement of food throughout the digestive system
- This is achieved by a rhythmic, muscular contraction of the walls of the tract which pushes the food in one direction



Digestive System - Component 2+5

Stomach

- A muscular bag like organ
- Walls are lined with mucous membrane
- Contains the cardiac sphincter at its connection with the oesophagus which prevents regurgitation



Digestive System - Component 2+5

Stomach Cont..

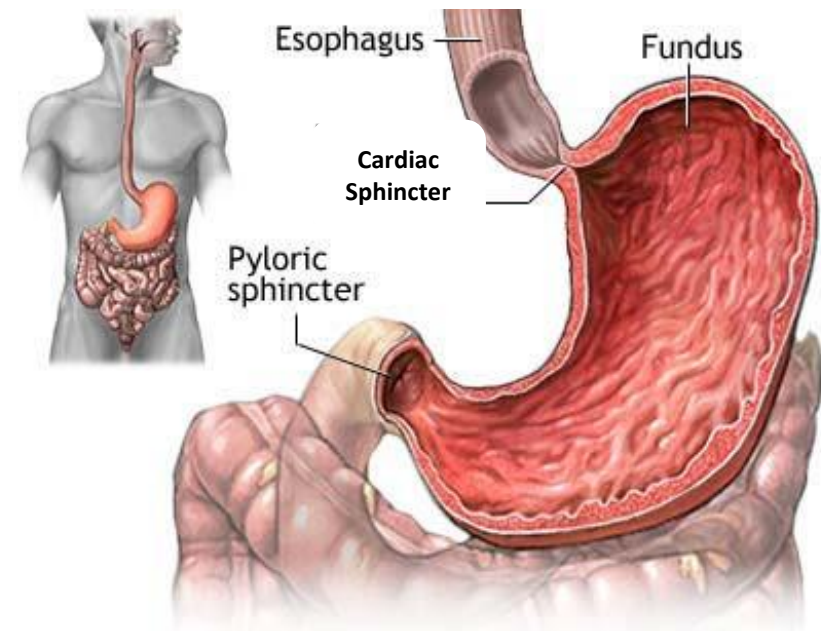
- **Mechanical Digestion**

The stomach is able to contract in a variety of ways to help churn food up

- **Chemical Digestion**

mixes food with gastric juices, hydrochloric acid and enzymes to liquefy

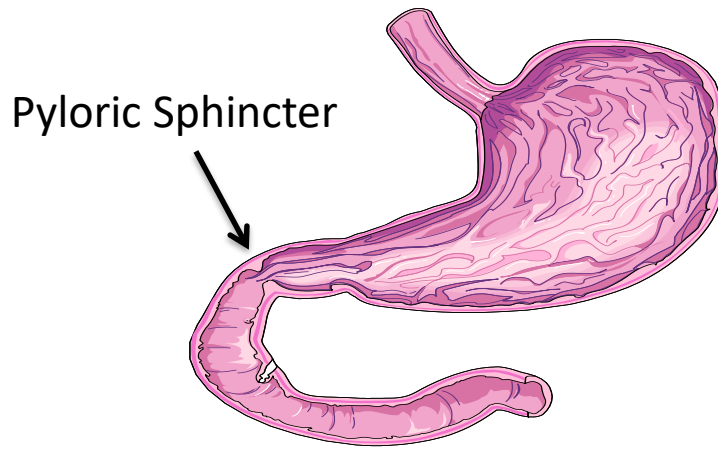
- Once food is liquefied in the stomach it turns into **Chyme**



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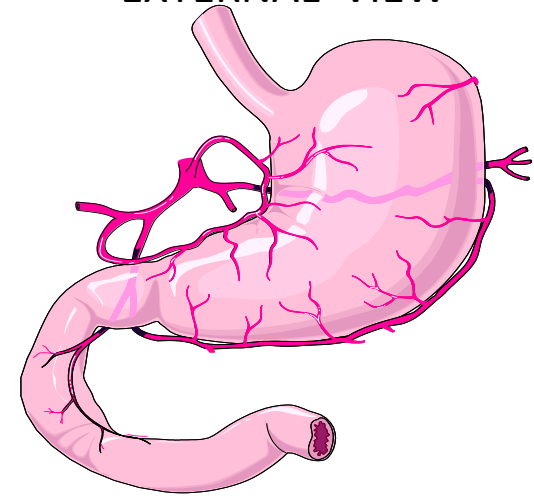
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Chyme then passes into the duodenum via the pyloric sphincter



INTERNAL VIEW

EXTERNAL VIEW

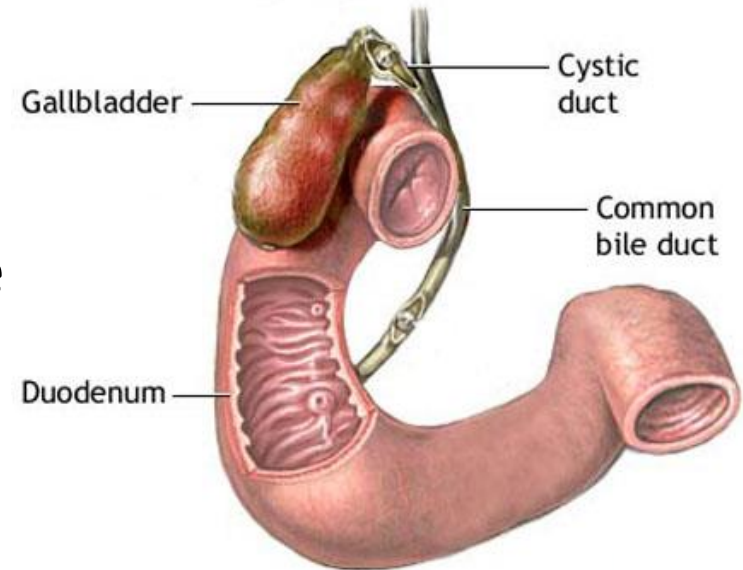


The sphincter helps prevent regurgitation back into the stomach from the duodenum

Digestive System - Component 2+5

Duodenum

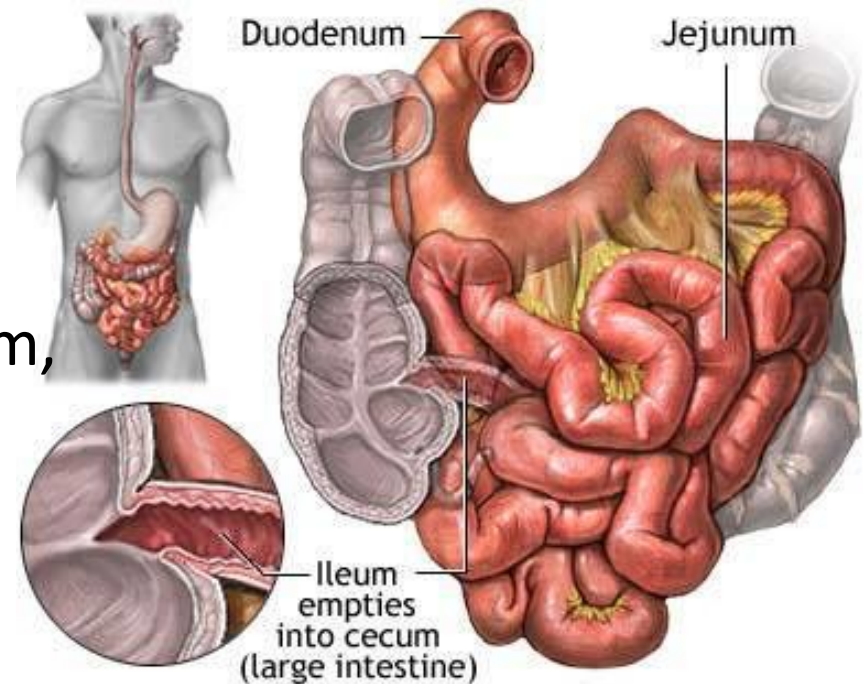
- The duodenum is around 25 cm in an adult and is the first part of the small intestine
- Both the bile duct and the pancreatic duct empty into the duodenum
- Chemical digestion of Carbohydrates, Proteins and fats occur across the small intestine



Digestive System - Component 2+5

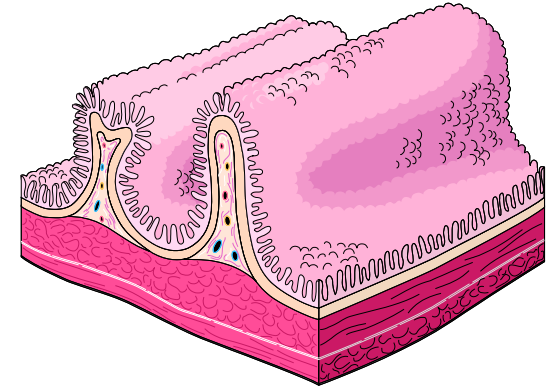
Small Intestine

- 5 metres long
- Muscular tube lined with connective tissue and mucous membrane
- Comprises of the Duodenum, Jejunum and the Ileum
- Continuous with the duodenum above and the large intestine below
- Most water absorption occurs here

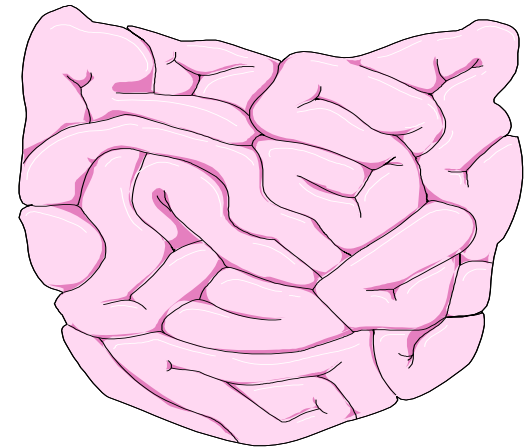


Digestive System - Component 2+5

- Outer covering is formed by the peritoneum
- Large surface area for digestion and absorption
- Nutrients pass through the inner lining and are absorbed by blood vessels contained in the wall of the intestine



VILLI

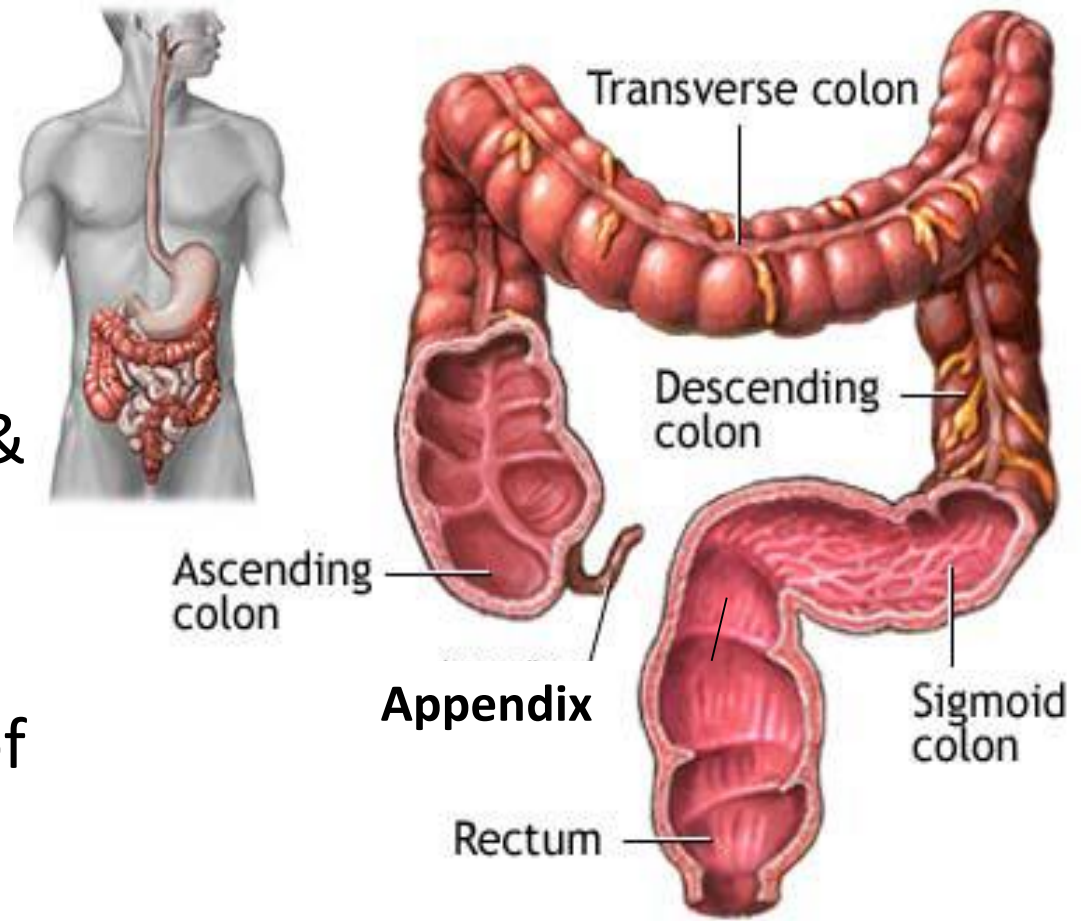


SMALL INTESTINE

Digestive System - Component 2+5

Large Intestine (Colon)

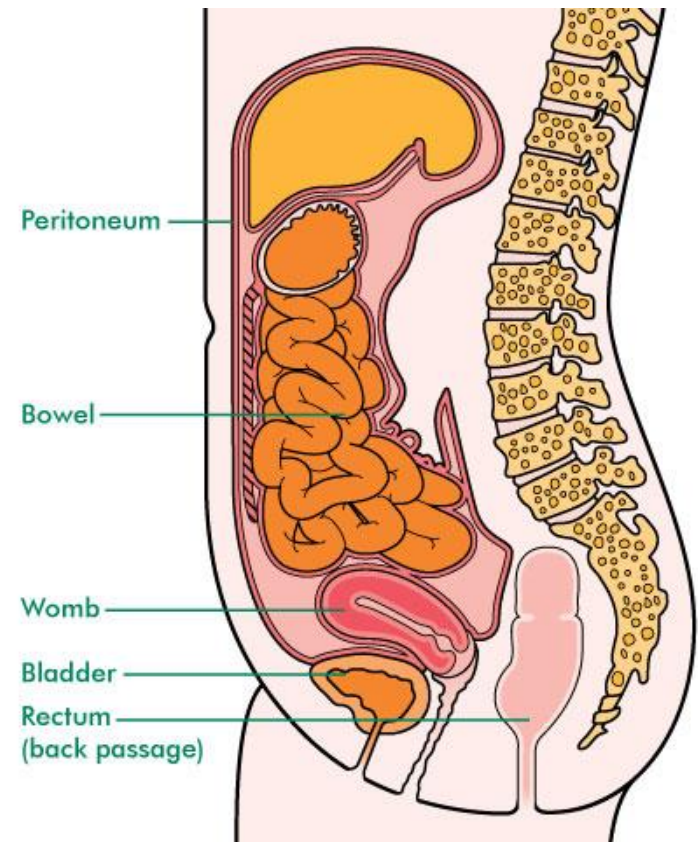
- 1.5m in length
- Absorbs electrolytes & vitamins
- Storage & expulsion of faeces



Digestive System - Component 2+5

Peritoneum

- Serous membrane that lines the abdominal cavity
- 2 Layers which cover abdominal organs
- Prevents friction with serous fluid
- Attaches organs to posterior abdominal wall
- Carries blood vessels that help fight infection





Digestive System - Component 2+5

Accessory Organs

These organs produce some of the chemicals necessary for digestion and include:

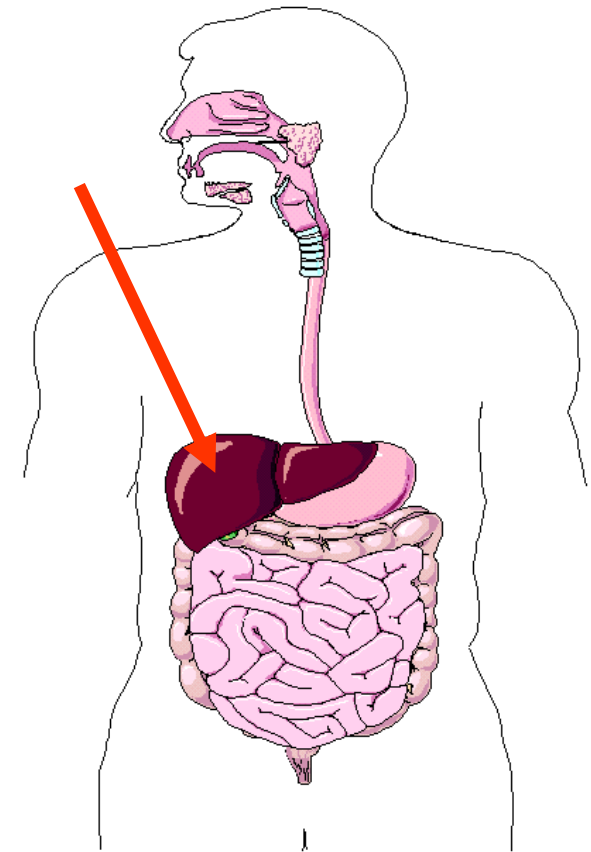
- Salivary glands
- Liver
- Pancreas

Digestive System - Component 2+5

Main Functions of Liver

- Carbohydrate, lipids(fat) and protein metabolism
- Processing of drugs and hormones
- Excretion of bilirubin
- Produces Bile
- Storage for vitamins and minerals
- Phagocytosis - breaks down cells
- Highest heat producing organ in body

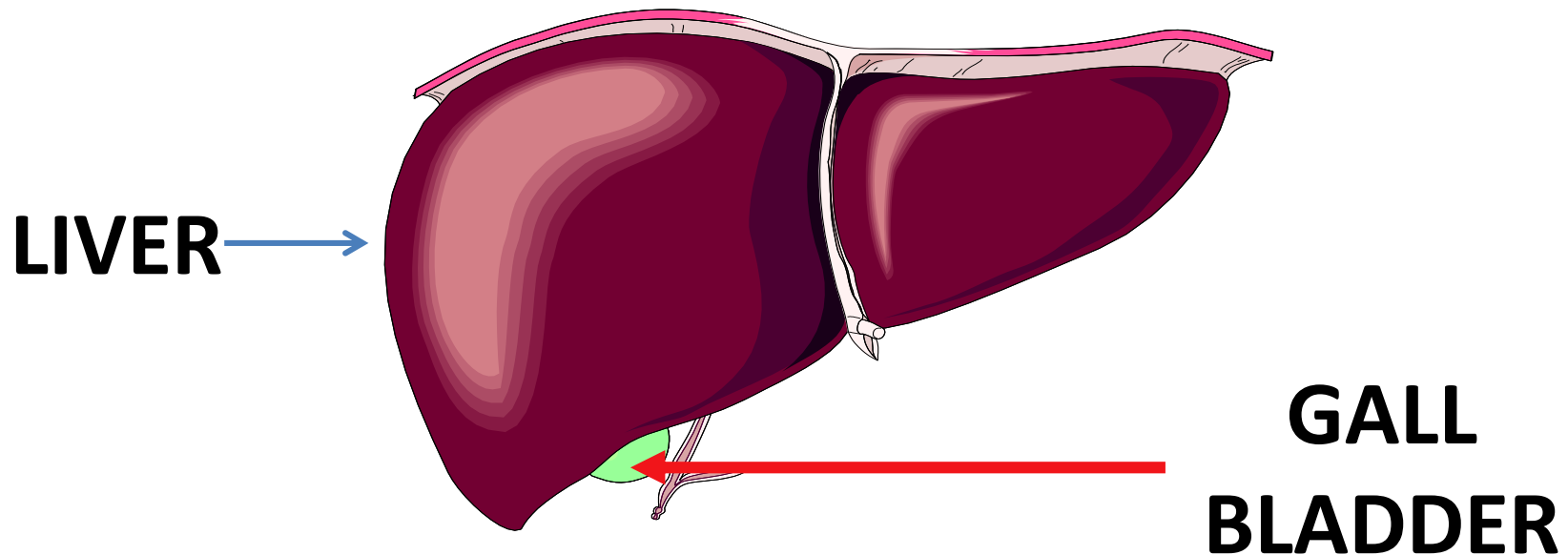
It makes, it breaks and it stores



Digestive System - Component 2+5

Gall Bladder & Bile

- Bile main function is to help digest fat to aid absorption
- Gall bladder stores and concentrates bile
- Releases Bile into the small intestine (1st part of Duodenum)





Digestive System - Component 2+5

PANCREAS

- Dual function organ
- 12-15cm long sitting behind the stomach
- **EXOCRINE** = Secretes **enzymes** (pancreatic juices) to aid with digestion
- **ENDOCRINE** = Secretes **hormones** (Insulin and Glucagon) to control blood sugar levels

Digestive System - Component 2+5



ANY QUESTIONS ???